



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to RAMESH AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

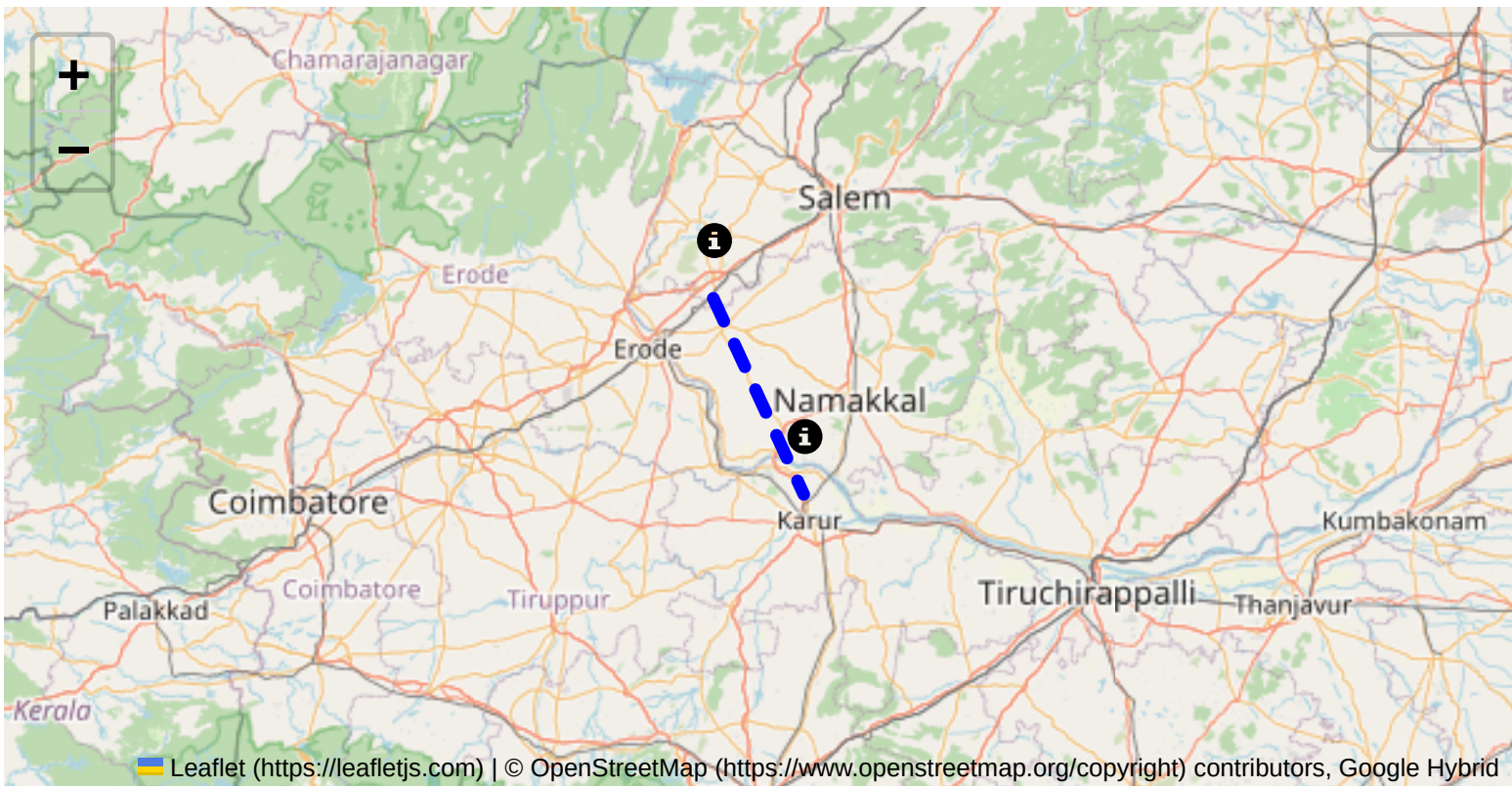
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

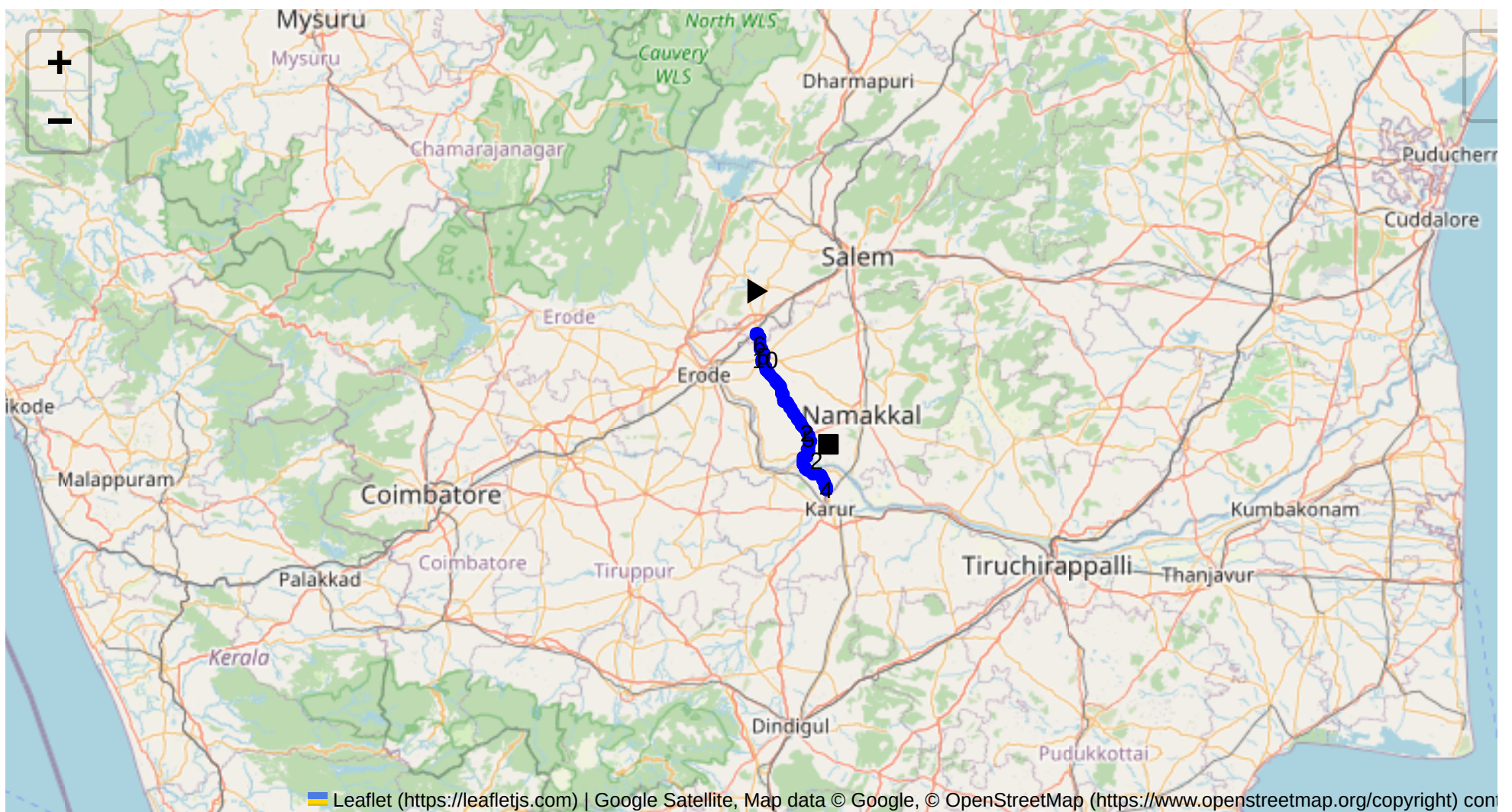
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 59.22 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours
Start: (11.4381, 77.8734)
End: (11.014291, 78.067088)



Welcome to the Journey Risk Management Study

Route Overview:

The route from CVQF+23W, Sangagiri, Tamil Nadu to 2378+VHG, Tamil Nadu spans approximately 59.22 kilometers and is primarily connected by state highways and local roads. The region includes agricultural areas, small towns, and rural landscapes.

1. Typical Weather Conditions:

- **Seasonal Changes:** Tamil Nadu experiences a tropical climate. The hottest months are March to May, with temperatures often exceeding 40°C. The rainy season is from June to September, with northeast monsoon causing heavy rains from October to December.
- **Weather Hazards:** Heavy rains can lead to waterlogged roads, reduced visibility, and increased risk of skidding. Summer heat can affect vehicle operation, potentially overheating engines and increasing tire wear.

2. Traffic Patterns:

- **Peak Hours:** Traffic tends to build up during morning (7:30-9:30 AM) and evening (5:30-7:30 PM) peak hours, particularly near towns and city exits.
- **Congestion Areas:** Key congestion points include market areas and major junctions in towns like Namakkal, where local traffic can slow down progress.

3. Road Quality and Infrastructure:

- **Road Conditions:** The primary roads are generally well-maintained. However, some rural sections may have narrow lanes without medians, adding risk during overtaking.
- **Infrastructure:** Bridges and overpasses are typically in good condition, but rural roads might lack proper lighting and signage.

4. Alternative Routes for Emergencies:

- Consider bypassing congested towns by using nearby rural roads, especially during peak hours or in case of construction or accidents. Using feeder roads parallel to the main highway can be useful in emergencies.

5. Local Regulations for Hazardous Material Transport:

- Transporting hazardous material requires adherence to state regulations, including obtaining necessary permits. Vehicles must display appropriate placards, and drivers should have the credentials required for handling hazardous materials.

6. Historical Incidents:

- Past incidents often involve overtaking on narrow roads, causing accidents with two-wheelers and smaller vehicles. No major spillages or hazardous material incidents have been documented on this specific route, but general caution is advised.

7. Environmental Considerations:

- Be aware of agricultural activities near the roads, as these areas may have people and animals crossing unexpectedly.

- No significant wildlife crossings are noted, but sensitivity to local ecology is important.

8. Communication Coverage:

- The route generally enjoys good mobile coverage, although some rural patches may experience weaker signals. Critical communication areas include small towns where coverage is reliable.

9. Emergency Response Times:

- Emergency response is estimated at 15-30 minutes in urban areas and 30-60 minutes in rural sections. Keep emergency contacts for police and medical services readily available.

Overall Summary of Risk Assessment:

Moderate to High Risk: The principal risks involve weather conditions, peak-hour traffic congestion, road quality in rural stretches, and regulatory compliance for hazardous materials. Pre-planning for alternative routes and maintaining vigilant speed and vehicle conditions can mitigate these risks effectively. Proper training on emergency procedures and regular maintenance checks for heavy vehicles transporting hazardous materials are essential.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	High	11.37868, 77.89498	15 KM/Hr
5	Turn	High	11.37850, 77.89453	15 KM/Hr
6	Turn	Medium	11.16755, 78.01568	30 KM/Hr
7	Turn	High	11.15573, 78.02176	15 KM/Hr
8	Turn	Medium	11.15565, 78.02171	30 KM/Hr
9	Turn	High	11.14545, 78.02208	15 KM/Hr
10	Blind Spot	Blind Spot	11.14480, 78.02291	10 KM/Hr
11	Blind Spot	Blind Spot	11.01166, 78.06877	10 KM/Hr
12	Turn	High	11.01161, 78.06848	15 KM/Hr
13	Turn	High	11.01423, 78.06745	15 KM/Hr
14	Turn	High	11.01421, 78.06733	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16755, 78.01568



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.15573, 78.02176



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.15565, 78.02171



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14545, 78.02208



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.01166, 78.06877



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.01161, 78.06848



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.01423, 78.06745



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.01421, 78.06733
